

Exact Minimum Field Partition for One-Step Prediction in a Finite Five-Agent ABCW Model

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Abstract

Which distinctions in a present state must be retained to predict a specified future exactly? We study this question on a finite reachable set of a deterministic five-agent ABCW model in which binary agent actions and strategies coevolve with a weighted directed influence field. The dataset is generated from four initial-field families and all 4,096 action-strategy initial conditions, yielding 56,536 transitions and 2,562 distinct current fields. We retain the current action and seek the coarsest field-only partition that uniquely determines the complete one-step-ahead field on every observed transition. Nine natural feature families and all 511 of their nonempty combinations provide strong approximate predictors but no exact nontrivial field compression. We therefore construct an incompatibility graph whose vertices are current fields and whose edges join fields that produce different next fields under a shared observed action. A proper 692-coloring gives a constructive upper bound. An independent anchor-action decomposition, followed by exact solution of the 50 nontrivial induced subgraphs, gives the matching lower bound. Hence the minimum number of field classes is exactly 692. This number applies only to the specified finite reachable set, field-only compression, retained current actions, one-step target, and exact-prediction requirement; it is not a universal macrostate count for ABCW systems.

Keywords: agent-based model; predictive partition; state aggregation; graph coloring; coarse-graining; exact prediction

Introduction

In complex dynamical systems, a more detailed description of the present does not necessarily provide a more useful description for predicting the future. A complete microscopic state may be sufficient to determine future evolution, but not every distinction encoded in that state need be relevant to a specified predictive task.

The central question of this study is therefore:

Which distinctions in the present must be retained in order to predict the future?

Equivalently:

How much of the present can be forgotten without losing the specified future?

This question is related to several established lines of research, including coarse-graining and state aggregation, lumpability in Markov chains, model minimization through bisimulation, causal states in computational mechanics, and state minimization for incompletely specified finite-state machines.

These approaches share a broad objective: to reduce a state description while preserving the distinctions required for dynamics or prediction. They differ, however, in what must be preserved and in the conditions imposed on the reduced representation.

The present study does not propose a new general theory encompassing these frameworks. Instead, it considers a finite and explicitly computable agent-based dynamical system and determines the minimum number of distinctions required to preserve a specified one-step prediction exactly.

The system studied here is a five-agent ABCW model. Each agent has a binary action and a binary strategy, while the influence relations among agents are represented by a time-dependent weight matrix W . Actions are updated by reference to local influence relations, and game outcomes feed back into both strategy adaptation and field evolution.

The payoff rule has a Minority-Game-type competitive structure. The ABCW model is not intended to reproduce the standard Minority Game itself. Rather, the rule is used to introduce an environment in which all agents cannot simultaneously succeed in the same sense. The relationship of this construction to the Minority Game and the El Farol problem is discussed in Section 2.

ABCW is therefore a finite agent system in which agent behavior and the interaction field coevolve. We represent the field by its deviation from the fully symmetric reference field W^* :

$$\Delta W = W - W^*.$$

For the five-agent model, we use four initial fields and exhaust all action and strategy initial conditions for each field, giving a total of 4,096 initial conditions. Each trajectory is followed until the complete state recurs. The resulting finite reachable dataset contains 56,536 transitions, 2,562 distinct current fields ΔW , and 11,202 distinct $(a, \Delta W)$ states.

Within this finite system, an observation is treated not merely as the selection of a variable but as a mapping

$$P : \Omega \longrightarrow Y,$$

which specifies which distinctions among complete states are retained and which are discarded.

A key distinction is that closure of the observed dynamics and prediction of a specified target are different requirements. For an observation P , requiring

$$P(X_t) \longrightarrow P(X_{t+1})$$

to be single-valued is a self-closure condition. By contrast, for a specified prediction target Z , requiring

$$P(X_t) \longrightarrow Z$$

to be single-valued is a target-prediction condition. The primary objective of this study is the latter.

Specifically, we retain the current action a_t and replace the current field ΔW_t with a coarser field-class label. We then ask whether this compressed description still determines the complete one-step-ahead field ΔW_{t+1} uniquely.

Because the object of compression is the field description, the action a_t is retained as an uncompressed conditioning variable. We do not simultaneously optimize a compression of the action space. This field-only restriction is part of the scope of the minimization problem solved in this paper.

Let

$$B : \Delta W \longrightarrow \mathcal{C}$$

be a field-only mapping, and define

$$P_B(X_t) = (a_t, B(\Delta W_t)).$$

We require

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

to be single-valued over all observed transitions. We call the ability of an observation to determine the specified target without ambiguity **predictive sufficiency**.

We first examine field compression based on human-designed network features, including norms, local structural descriptors, and out-strength. Some of these features achieve high predictive performance. Within the candidate family tested, however, no feature combination both merges genuinely distinct values of ΔW and predicts the complete one-step-ahead field exactly.

This failure does not imply that exact prediction requires all 2,562 observed fields to remain distinct. It shows only that the tested feature family does not identify an exact nontrivial compression. We therefore remove the restriction that the partition must be specified in advance through human-designed features and instead derive the necessary distinctions directly from the observed dynamics.

The finite dataset can be read as a partially specified input-output relation

$$(\Delta W, a) \longmapsto \Delta W',$$

where ΔW is the current field, a is the action condition, and $\Delta W'$ is the complete one-step-ahead field. If two current fields produce different next fields under a common action condition, they cannot belong to the same field class.

Accordingly, we construct an incompatibility graph G whose 2,562 vertices are the distinct observed current fields. Two fields are connected by an edge if, under at least one action condition observed for both fields, they produce different next fields. A field-only partition that preserves exact one-step prediction is then equivalent to a proper coloring of G .

The reduced field-class label is not required to serve recursively as the internal state at the next time step. The target being preserved is the complete next field ΔW_{t+1} , rather than the successor class label. Consequently, the recursive successor-closure condition encountered in classical minimization of incompletely specified finite-state machines is not imposed here. For this depth-1 prediction problem, the minimum number of field classes is therefore given directly by

$$|\text{Im } B_{\min}| = \chi(G).$$

Compatibility need not itself be assumed to form an equivalence relation. Pairwise distinctions that cannot be merged are encoded directly as incompatibility edges, and a proper coloring separates every incompatible pair. The relationship between this formulation and classical state-minimization problems is developed in Section 2.

The main result is

$$\chi(G) = 692.$$

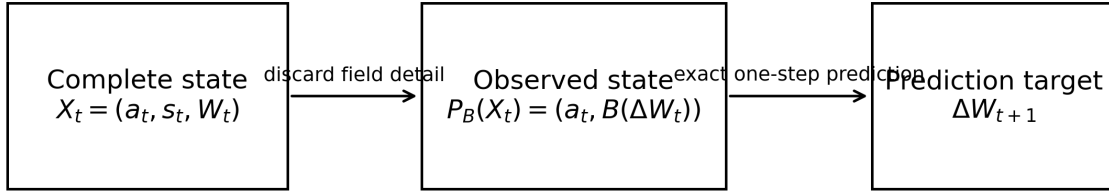
Thus, for the finite reachable set analyzed here and for the specified task of exact one-step field prediction, the 2,562 distinct current fields can be compressed to 692 field classes:

$$2562 \longrightarrow 692,$$

while preserving exact prediction of the complete next field.

Figure 1 summarizes the predictive-partition problem and the resulting reduction.

Figure 1. Predictive-partition problem



Field partition: 2,562 observed fields $\rightarrow$ minimum 692 classes

Which distinctions in the present field are necessary for the specified future?

Figure 1: Figure 1. Predictive-partition problem. The complete ABCW state is compressed by retaining the action and replacing the full field deviation with a field-only partition label. The prediction target is the exact one-step-ahead field deviation. The main result reduces 2,562 observed fields to a minimum of 692 classes.

Figure 1. Predictive-partition problem. The complete ABCW state is compressed by retaining the action and replacing the full field deviation with a field-only partition label. The prediction target is the exact one-step-ahead field deviation. The main result reduces 2,562 observed fields to a minimum of 692 classes.

Measured by the number of field classes, the corresponding compression rate is

$$1 - \frac{692}{2562} \simeq 72.99\%.$$

This is a reduction in the number of field classes; it should not be interpreted as a 72.99% reduction in information content.

The value 692 is not merely the output of a coloring heuristic. A lower bound is obtained by fixing the anchor action

$$a = (-, -, -, -, -).$$

Under this action, the 1,239 observed current fields produce 625 distinct next fields, so at least 625 field classes are required. Within 50 of these 625 anchor groups, additional incompatibilities appear when other action conditions are taken into account. The chromatic numbers of the corresponding induced subgraphs are solved exactly by combining clique lower bounds with colorability searches. These refinements require 67 additional classes, giving the lower bound

$$625 + 67 = 692.$$

For the matching upper bound, we explicitly construct a proper 692-coloring of the incompatibility graph. Mapping this coloring back to all 56,536 observed transitions verifies that every observed pair $(a_t, B(\Delta W_t))$ determines a unique ΔW_{t+1} . The upper and lower bounds therefore coincide, establishing $\chi(G) = 692$ exactly.

This result does not imply that 692 is a universal number of macrostates for ABCW systems. It depends on the five-agent model, the four initial-field families—Baseline, Hub, Local, and Hub+Local—the finite reachable set generated from them, the retention of action as an uncompressed conditioning variable, the field-only restriction, and the requirement of exact prediction of the complete one-step-ahead field. We therefore do not claim that 692 is the minimum state count of ABCW in general.

The broader point is that microscopic distinctness and predictive relevance need not coincide. In this finite ABCW system, fields that are microscopically different can sometimes be assigned to the same predictive class without losing the specified future. At the same time, simple network features and geometric proximity between fields do not fully identify which distinctions must be preserved.

The resulting partition occupies an intermediate descriptive level between a complete microscopic representation and a small set of simple macroscopic statistics: it retains exactly the distinctions required by the specified prediction task on the observed finite system. This does not by itself make the 692 classes an interpretable set of macroscopic variables, but it provides an exact target against which candidate coarse descriptions can be evaluated.

The contributions of this study are threefold.

First, for a finite ABCW system, we distinguish self-closure from target prediction and formulate the question of which present distinctions must be retained as an explicit observation-minimization problem.

Second, we distinguish compression by preselected network features from a predictive partition derived from future responses, and we formulate the resulting field-only, action-conditioned, depth-1 problem as a graph-coloring problem.

Third, for the finite reachable set considered here, we establish that the exact minimum number of field classes is 692 by providing both a constructive upper bound and an independent matching lower bound. The result demonstrates that not all 2,562 microscopic field distinctions are required for exact one-step field prediction.

These contributions do not assert novelty for the general theories of state minimization, lumpability, bisimulation, or causal states. Rather, the study formulates a specifically scoped prediction problem for a finite coevolving agent-field system and solves its exact minimum on a fully inspectable dataset.

The remainder of the paper is organized as follows. Section 2 positions the study in relation to the El Farol problem, the Minority Game, state aggregation, lumpability, bisimulation, computational mechanics, and minimization of incompletely specified finite-state machines. Section 3 defines the ABCW model and the finite reachable dataset. Section 4 formalizes observation, self-closure, predictive sufficiency, and the prediction target used in this paper. Section 5 evaluates compression based on natural network features. Section 6 constructs the incompatibility graph and determines the exact minimum field-only partition. Section 7 analyzes the internal structure of the 692 classes. Section 8 discusses the interpretation, scope, and limitations of the result. Section 9 concludes the paper.

Related Work and Positioning

Competitive agent models: El Farol and the Minority Game

The competitive payoff structure of the ABCW model shares part of its motivation with the El Farol Bar problem and the Minority Game.

The El Farol Bar problem introduced by Arthur (1994) considers a setting in which several agents seek access to the same limited opportunity or resource, so that the desirability of an action depends on the choices of other agents. Attendance is advantageous when relatively few agents attend, but its value declines when too many make the same choice. Consequently, the rationality of an action cannot be fixed independently of the behavior of the population.

The Minority Game introduced by Challet and Zhang (1997) formulates this competitive structure as a simplified binary-choice game. At each time step, agents select one of two alternatives, and agents in the minority receive a payoff. The value of an action is therefore not intrinsic to that action but depends on how many other agents choose it.

ABCW employs this Minority-Game-type competition. Each of five agents takes a binary action, with agents in the minority designated as winners and those in the majority as losers. This creates a competitive payoff environment in which all agents cannot simultaneously succeed in the same sense.

ABCW is not intended, however, to reproduce or extend the standard Minority Game, nor is it designed to analyze its characteristic collective phenomena. Standard Minority Game research commonly focuses on agents' strategy sets, information histories, adaptation, collective efficiency, and aggregate fluctuations. By contrast, the central object in the present study is the time-dependent field W representing influence relations among agents.

In ABCW, game outcomes feed back not only into agent scores but also into the evolution of the field. The Minority-Game-type rule is therefore a component that generates the coupled process

$$\text{action} \longrightarrow \text{outcome} \longrightarrow \text{field update} \longrightarrow \text{next action},$$

rather than the final object of analysis. The primary question is not which strategy dominates in a Minority Game or how efficiently the population coordinates. Instead, for the field dynamics generated by this competitive agent system, we ask:

Which distinctions among current fields must be retained to predict the complete one-step-ahead field exactly?

The El Farol problem and the Minority Game thus provide the agent-modeling background for the competitive dynamics of ABCW. They are not the direct theoretical basis of the predictive state-reduction problem addressed here. The more immediate connections are to state aggregation and finite-state minimization.

State aggregation and the question of relevant distinctions

The central question of this study—which distinctions in the present must be retained for prediction—belongs to a broad class of problems concerned with aggregating a state space while preserving properties required for dynamics or prediction.

A classical example is lumpability in Markov chains. In the formulation of Kemeny and Snell (1960), a partition of the state space into lumps yields a closed Markov process on the aggregated states when, for any two states in the same lump, the transition probabilities into every destination lump coincide. Lumpability therefore provides conditions under which microscopic states can be merged while retaining well-defined reduced dynamics.

The self-closure condition introduced in Section 4 has a related motivation in deterministic finite dynamics. If

$$P(X) = P(X')$$

implies

$$P(\Phi(X)) = P(\Phi(X')),$$

then two complete states assigned to the same observed state also evolve to the same observed state at the next time step. Here X is a complete state, Φ is the one-step update map on complete states, and $P : \Omega \rightarrow Y$ is a general observation map on the complete state space. This general map P should be distinguished from the field-only map $B : \Delta W \rightarrow \mathcal{C}$ that is later minimized.

Self-closure is not, however, the primary requirement in this paper. We distinguish prediction of the next observed state,

$$P(X_{t+1}),$$

from prediction of a separately specified target,

$$Z = g(X_{t+1}).$$

The required condition is therefore not necessarily closure of

$$P(X_t) \longrightarrow P(X_{t+1}),$$

but single-valuedness of

$$P(X_t) \longrightarrow Z.$$

This distinction between closed reduced dynamics and sufficient information for a specified target is the starting point of our state-aggregation problem.

Bisimulation and model minimization

The idea of merging states while preserving future behavior is also closely related to bisimulation and bisimulation-based model minimization.

Givan, Dean, and Greig (2003), for example, study equivalence notions for Markov decision processes and use bisimulation-based aggregation to construct reduced MDPs that preserve optimal policies of the original process. The common principle is that states are grouped not because they are superficially similar, but because their distinction is unnecessary for the future behavior that must be preserved.

The system and preservation target considered here are different. ABCW is not an MDP; on the stored finite reachable set, it has deterministic update rules. Nor do we minimize the complete ABCW state. Writing the complete state as

$$X = (a, s, W),$$

we retain the action a in the observation and compress only the field deviation ΔW through

$$B : \Delta W \longrightarrow \mathcal{C}.$$

Moreover, the preserved target is neither an optimal policy nor a reward structure nor the full reduced dynamics. It is the complete one-step-ahead field ΔW_{t+1} . We therefore do not describe the resulting minimum partition as a bisimulation quotient.

Computational mechanics and causal states

Computational mechanics provides another closely related predictive perspective. Shalizi and Crutchfield (2001) define causal states by identifying past histories that induce the same conditional probability distribution over futures. The resulting ϵ -machine is characterized as a minimal representation retaining the information required for prediction.

The conceptual connection to the present study is clear: distinctions among complete histories or microscopic states are retained only when they matter for the future. Similarly, the 692 classes

derived here are defined not by static similarity in network structure but by the one-step field responses generated under action conditions.

The two constructions should nevertheless not be identified. Causal states are, in principle, equivalence classes of histories defined by conditional distributions over future sequences. The present study instead partitions current fields ΔW in a finite reachable set, with the prediction target restricted to the complete one-step-ahead field ΔW_{t+1} .

The ABCW data are also partially specified: not every action condition is observed for every current field. As a result, the relation “no conflict occurs under the observed common conditions” need not be a transitive equivalence relation. We therefore do not call the 692 classes causal states or an ϵ -machine. The relevant connection is the shared objective of preserving future-relevant distinctions.

Incompletely specified finite-state machine minimization

The most direct mathematical correspondence is with minimization of incompletely specified finite-state machines. A classical treatment is given by Paull and Unger (1959). When outputs or transitions are not specified for every state-input pair, state equivalence for a fully specified machine cannot be applied directly. Minimization methods instead examine whether states are mutually compatible and construct compatible classes that permit a reduction in the number of states. This remains a standard state-reduction problem in finite-state machine theory (Paull and Unger 1959; Kohavi and Jha 2009).

In classical minimization of an incompletely specified machine, compatibility of current input-output behavior is generally not sufficient. If the reduced machine is to operate recursively, successor states must also be consistent with the selected reduced classes. The minimization problem therefore involves a cover satisfying both compatibility and closure conditions.

The ABCW dataset has a related structure when viewed as partially specified input-output behavior. Taking ΔW as the current field, a as an input condition, and $\Delta W'$ as the complete next-field output, the observations define a partial table

$$(\Delta W, a) \mapsto \Delta W'.$$

Suppose that two current fields ΔW_i and ΔW_j are both observed under the same action condition a but generate different next fields,

$$\Delta W'_i \neq \Delta W'_j.$$

The two current fields cannot then be assigned to the same field class. We represent this constraint by an incompatibility graph whose 2,562 vertices are the distinct observed current fields and whose edges join pairs that produce different next fields under at least one common observed action condition.

Adjacent vertices cannot share a field class. A field-only partition preserving exact one-step prediction therefore gives a proper coloring of the incompatibility graph. Conversely, in a proper coloring, two fields assigned the same color have no observed common action condition under which they produce conflicting next fields. Treating each color as a field class consequently preserves the single-valued relation

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}.$$

The minimum number of classes is therefore exactly

$$|\text{Im } B_{\min}| = \chi(G).$$

An important difference from classical incompletely specified FSM minimization is that the field-class label is not required to function recursively as the internal state at the next time step. The object preserved here is the complete next field ΔW_{t+1} , not a successor class label. The requirement is only that the depth-1 target-prediction relation

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

be single-valued. No recursive closure condition on successor classes is therefore required. For the problem defined in this paper, minimization reduces directly to proper coloring rather than to a closed-cover construction.

Nor must compatibility itself be assumed to be an equivalence relation. In partially specified data, the absence of an observed conflict need not be transitive. We instead encode every pair that cannot be merged as an incompatibility edge and enforce the required pairwise constraints through proper coloring.

This simplification is not presented as a new general minimization theory for incompletely specified finite-state machines. It is a consequence of the specific preservation target adopted here: because the ABCW problem is restricted to an external one-step target, the classical recursive closure requirement is unnecessary, and the minimum partition is expressible as a pure graph-coloring problem.

Feature-based coarse-graining and predictive partitioning

We further distinguish two approaches to state compression.

The first computes a set of preselected features from the state:

$$\Delta W \longrightarrow B_{\text{feature}}(\Delta W).$$

Examples considered here include norms, local structural descriptors, hub structure, and out-strength. The advantage of this approach is interpretability: the resulting observables have meanings chosen in advance. A low-dimensional feature representation, however, does not necessarily constitute genuine coarse-graining. On a finite state set, even a low-dimensional feature vector may uniquely identify every microscopic state. When features do merge states, they may also discard distinctions required for prediction.

The second approach does not choose the features in advance. Instead, it derives the required distinctions from future responses, conceptually through the response structure

$$\Delta W \longrightarrow (a \longmapsto \Delta W').$$

The 692-class partition belongs to this second approach. Its classes are not guaranteed to admit an immediate interpretation in terms of simple network statistics. In return, the partition is determined directly by the requirement that the specified predictive target be preserved.

We refer to this property as **predictive sufficiency**. An observation P is predictively sufficient for a target Z on the dataset under consideration if

$$P(X_t) \longrightarrow Z$$

is single-valued, so that Z is determined without ambiguity by $P(X_t)$. Predictive sufficiency does not mean merely that a state can be expressed with a small number of variables. The question is instead how many distinctions can be discarded while retaining all distinctions required for the specified target.

Position of the present study

The present study has two distinct connections to previous work.

The first concerns the competitive agent dynamics of ABCW. The model shares with the El Farol problem and the Minority Game a payoff structure in which an agent’s outcome depends on the choices of other agents and all agents cannot simultaneously succeed in the same sense. We do not analyze standard Minority Game phenomena or strategy adaptation as ends in themselves. The competitive rule is used to generate coupled evolution between agent behavior and the field W .

The second connection concerns the theoretical positioning of the predictive state-reduction problem. With lumpability, we share the concern of obtaining a well-defined description after aggregation, but we distinguish self-closure from prediction of a separately specified target. With bisimulation and MDP minimization, we share the principle of aggregation according to future behavior, but we do not preserve optimal policies or a complete reduced MDP. With computational mechanics, we share the objective of retaining future-relevant distinctions, but the 692 classes are not history-based causal states: they preserve one-step, action-conditioned field responses on a finite reachable set. With incompletely specified FSM minimization, the correspondence is more direct because both problems involve compatibility and incompatibility in partially specified input-output behavior; our target restriction makes recursive closure unnecessary.

The most limited characterization of the present study is therefore:

It is a concrete state-minimization problem that determines the exact minimum field-only partition preserving a specified one-step prediction for partially specified input-output data generated by a finite competitive agent system.

We do not locate the novelty of the study in the general ideas of the Minority Game, state minimization, or predictive representation themselves. The specific contribution is to formulate the depth-1 prediction problem explicitly for the observed finite reachable set of the five-agent ABCW model and to establish, for its incompatibility graph, that

$$\chi(G) = 692$$

by a constructive upper bound and an independent matching lower bound. Consequently, the complete one-step-ahead field can be predicted exactly without retaining all 2,562 microscopically distinct current fields, whereas no partition into 691 or fewer field classes can preserve that prediction.

This finite and fully inspectable example makes explicit the difference between being distinct as a state and needing to remain distinct for a specified prediction.

ABCW Model and Dataset

This section defines the five-agent ABCW model analyzed in this paper and the finite reachable dataset used in the predictive-partition problem. The purpose is not to reconstruct the exploratory development of ABCW, but to fix the state variables, update rules, field representation, and scope of the subsequent analysis.

State Variables and Interaction Network

Let the set of agents be

$$V = \{1, \dots, n\},$$

with

$$n = 5$$

throughout this paper. At time t , each agent i has a binary action

$$a_i(t) \in \{-1, +1\}$$

and a binary strategy

$$s_i(t) \in \{-1, +1\}.$$

The value $s_i(t) = +1$ denotes a trend-following strategy that follows the reference signal, whereas $s_i(t) = -1$ denotes a contrarian strategy that acts against it.

Influence relations among agents are represented by a fixed topology

$$E = (E_{ij}), \quad E_{ij} \in \{0, 1\}$$

and a time-dependent nonnegative weight matrix

$$W(t) = (w_{ij}(t)).$$

The condition $E_{ij} = 1$ indicates that the directed edge $i \rightarrow j$ is permitted by the topology, and $w_{ij}(t)$ represents the strength of the influence exerted by agent i on agent j at time t . If its weight reaches zero, the topological edge itself is not removed as long as $E_{ij} = 1$.

Because strategy is included as a state variable, the complete state is defined as

$$X(t) = (a(t), s(t), W(t)).$$

Table 1 summarizes the principal variables and their roles.

Table 1. ABCW variables and update-rule summary

Symbol	Meaning	Domain / type	Role / update
$a_i(t)$	Action	$\{-1, +1\}$	Current binary action
$s_i(t)$	Strategy	$\{-1, +1\}$	Trend-following / contrarian; loser flips strategy
$u_i(t)$	Payoff	$\{-1, +1\}$ for $n = 5$	Minority +1, majority −1; all-same gives −1
$W(t)$	Influence field	Nonnegative weighted directed matrix	Outgoing edges updated by source payoff
$\sigma_i(t)$	Reference signal	$[-1, 1]$	Mean action among strongest effective incoming references
$\Delta W(t)$	Field deviation	Matrix	$W(t) - W^*$

Payoff, Reference Signal, and Update Rules

Minority-game payoff

At each time step, every agent chooses either -1 or $+1$. Agents in the minority are designated winners and those in the majority losers. The payoff is

$$u_i(t) \in \{-1, +1\},$$

with $u_i(t) = +1$ for a minority agent and $u_i(t) = -1$ for a majority agent. Because $n = 5$ is odd, a tie between the two binary actions is impossible. If all agents choose the same action, all are treated as losers and receive $u_i(t) = -1$.

This payoff rule is not intended to reproduce the full standard Minority Game. It is used to introduce into ABCW a competitive condition under which all agents cannot win simultaneously.

Reference signal

The set of agents that agent i can reference is

$$\mathcal{N}_i = \{j \in V \mid E_{ji} = 1\}.$$

Define the maximum incoming weight available to agent i as

$$m_i(t) = \begin{cases} \max_{j \in \mathcal{N}_i} w_{ji}(t), & \mathcal{N}_i \neq \emptyset, \\ 0, & \mathcal{N}_i = \emptyset \end{cases}.$$

When $m_i(t) > 0$, let

$$M_i(t) = \{j \in \mathcal{N}_i \mid w_{ji}(t) = m_i(t)\}$$

be the set of agents attaining that maximum, and define the reference signal as

$$\sigma_i(t) = \frac{1}{|M_i(t)|} \sum_{j \in M_i(t)} a_j(t).$$

If several agents share the maximum incoming weight, no single agent is selected arbitrarily; all tied agents are referenced with equal weight. If their actions balance, then $\sigma_i(t) = 0$. We also set $\sigma_i(t) = 0$ when no referenceable agent exists or when the maximum available influence is zero, treating both cases as the absence of an effective reference signal.

The game outcome $u_i(t)$ is thus determined by the actions of all agents, whereas the reference information $\sigma_i(t)$ used for the next action is determined by local influence relations.

Strategy adaptation

In the adaptive ABCW model used here, each agent updates its strategy using only the outcome of the immediately preceding game. A winner retains its strategy, whereas a loser reverses it. Thus,

$$s_i(t+1) = \begin{cases} s_i(t), & u_i(t) = +1, \\ -s_i(t), & u_i(t) = -1 \end{cases}$$

or, equivalently,

$$s_i(t+1) = u_i(t)s_i(t).$$

Action update

After payoffs have been calculated from the current actions and reference signals from the current field, strategies are updated first. A losing agent uses this **updated strategy**, rather than its previous strategy, to determine its next action. The action update is therefore

$$a_i(t+1) = \begin{cases} a_i(t), & u_i(t) = +1, \\ a_i(t), & u_i(t) = -1, \sigma_i(t) = 0, \\ \text{sgn}(s_i(t+1)\sigma_i(t)), & u_i(t) = -1, \sigma_i(t) \neq 0 \end{cases}.$$

A winner retains its current action. A loser also retains its current action if no effective reference signal is available; otherwise, it selects the next action according to its updated strategy and the reference signal.

Field update

Finally, the payoff is fed back into the field. Let the learning rate be $\eta > 0$. All experiments analyzed in this paper use

$$\eta = 1.$$

The field update is

$$w_{ij}(t+1) = \begin{cases} \max(0, w_{ij}(t) + \eta u_i(t)), & E_{ij} = 1, \\ 0, & E_{ij} = 0 \end{cases}.$$

Thus, when agent i wins, the weights of the existing outgoing edges from that agent increase; when it loses, they decrease, subject to the lower bound of zero.

The update order for one game is fixed as

$$\begin{array}{c} (a(t), s(t), W(t)) \\ \downarrow \\ u(t), \sigma(t) \\ \downarrow \\ s(t+1) \\ \downarrow \\ a(t+1) \\ \downarrow \\ W(t+1) \end{array}.$$

This ordering is part of the model definition because losing agents use their updated, rather than previous, strategies when selecting their next actions.

Every branch of the payoff, reference, strategy, action, and field updates is resolved by these rules. In particular, tied reference agents are averaged rather than selected arbitrarily, a zero reference signal invokes the action-retention rule above, and no random tie-breaking or stochastic choice is used. Consequently, for fixed $(a(t), s(t), W(t))$ and E , the complete successor state is unique. The ABCW system analyzed here is therefore deterministic, and repeated runs from the same initial state generate the same finite trajectory and transition records.

Figure 2 summarizes the complete update cycle used to generate the dataset.

Figure 2. ABCW update cycle. Update order for the adaptive five-agent ABCW model used to construct the dataset. Payoffs and reference signals are computed from the current state; strategies are updated, then actions, then the influence field.

Field Representation

For the five-agent system, define the fully symmetric reference field as

$$W^* = \begin{array}{ccccc} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{array}.$$

Figure 2. ABCW update cycle

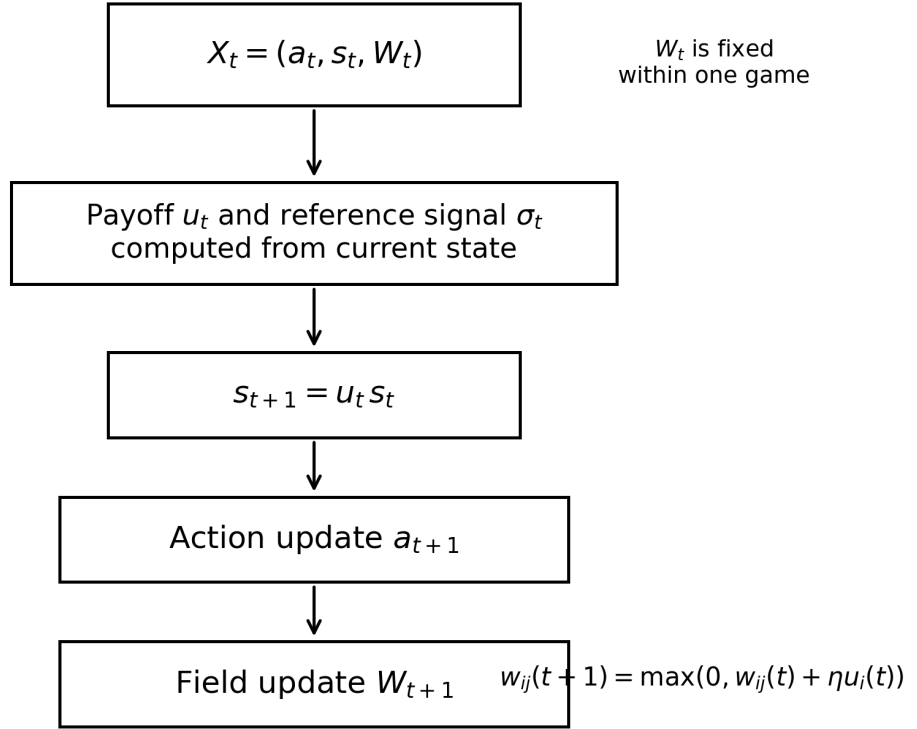


Figure 2: Figure 2. ABCW update cycle. Update order for the adaptive five-agent ABCW model used to construct the dataset. Payoffs and reference signals are computed from the current state; strategies are updated, then actions, then the influence field.

The reference field W^* is not assumed to be an equilibrium or a naturally privileged social state. It serves only as a reference point for describing field deformation.

The field deviation is defined as

$$\Delta W(t) = W(t) - W^*,$$

or componentwise as

$$\Delta w_{ij}(t) = w_{ij}(t) - w_{ij}^*.$$

Thus, $\Delta W(t) = 0$ means that the current field coincides with the reference field. A nonzero deviation $\Delta W \neq 0$ does not necessarily imply matrix asymmetry $W \neq W^\top$. Throughout this paper, *field deviation* refers to the difference from the reference field itself. This ΔW is the field variable used in the subsequent prediction problem.

Dataset Construction

The analysis uses four initial fields for the five-agent ABCW model:

- Baseline
- Hub
- Local
- Hub+Local

For Baseline, $W(0) = W^*$. In each of the other three initial fields, four directed edges whose reference weight is 1 are strengthened to 4, while the update rules remain unchanged. Only the placement of these strong edges differs. Numbering the agents $1, \dots, 5$ and denoting the directed edge $i \rightarrow j$ by (i, j) , the strengthened edges are:

- Hub: $(1, 2), (1, 3), (1, 4), (1, 5)$
- Local: $(1, 2), (2, 3), (3, 4), (4, 5)$
- Hub+Local: $(1, 3), (1, 5), (2, 3), (4, 5)$

In Hub, the four outgoing edges from agent 1 to the other agents are strengthened. Local strengthens a chain of four edges. Hub+Local combines two outgoing edges from agent 1 with two local competing edges that share their destination agents.

Each of these three non-baseline fields therefore has four nonzero initial deviation components and the same Frobenius norm,

$$\|\Delta W(0)\|_F = \sqrt{4 \times 3^2} = 6.$$

These configurations are not intended as an exhaustive classification of network structures. They serve as initial-condition families from which a finite reachable state set is constructed.

For each initial field, we exhaust all combinations of the five-agent initial action vector $a(0)$ and initial strategy vector $s(0)$. Because every component is binary, this gives

$$2^5 \times 2^5 = 1024$$

initial conditions per field and

$$4 \times 1024 = 4096$$

initial conditions in total.

Each case is evolved until the complete state recurs, and every transition encountered before recurrence is collected. This procedure yields 56,536 transitions containing 2,562 distinct current fields ΔW and 11,202 distinct $(a, \Delta W)$ states.

Table 2 summarizes the dataset construction and the resulting counts.

Table 2. Dataset construction and summary

Quantity	Value
Agents	5
Learning rate η	1
Initial field configurations	4
Action configurations per field	32
Strategy configurations per field	32
Initial conditions	4096
Observed transitions	56536
Distinct current fields	2562
Distinct $(a, \Delta W)$ states	11202

The 56,536 transitions do not represent the complete theoretical state space of ABCW. All subsequent results should be interpreted as statements about the stored finite transition set reached from the four initial fields and 4,096 initial conditions defined above. We make no claim of generalization to unreachable states, other numbers of agents, or other initial fields. Nor do we interpret the recurrence of every trajectory among the 4,096 cases as a proof that the weights are bounded in general.

Observation and Predictive Objective

This section formulates, for the finite ABCW system defined above, the problem of determining which distinctions in the present must be retained to predict a specified future exactly. The central point is to distinguish closure of the observed dynamics from sufficiency for a particular prediction target.

Observation as Information Loss

Let Ω denote the complete state space, and represent an observation as a mapping

$$P : \Omega \longrightarrow Y.$$

If two distinct complete states $X, X' \in \Omega$ satisfy

$$P(X) = P(X'),$$

then the observation P discards the distinction between them and treats them as the same observed state. An observation is therefore not merely a choice of which variables to inspect. It is a mapping that determines which distinctions among complete states are retained and which are identified.

The complete ABCW state is

$$X = (a, s, W).$$

Because the reference field W^* is fixed, W and ΔW are in one-to-one correspondence. For example,

$$P_W(X) = \Delta W$$

discards distinctions in action a and strategy s , whereas

$$P_{aW}(X) = (a, \Delta W)$$

discards only distinctions in strategy s .

Self-Closure and Target Prediction

Write the deterministic ABCW update as

$$X_{t+1} = \Phi(X_t).$$

An observation P is called **self-closed** if

$$P(X) = P(X')$$

always implies

$$P(\Phi(X)) = P(\Phi(X')).$$

Under this condition, the next observed state is uniquely determined by the current observed state. Self-closure is a deterministic quotient-consistency condition that makes the coarse-grained transition well defined. It is related to aggregation conditions in lumpability and bisimulation, but it is not identical to causal-state construction from conditional distributions over future sequences (Shalizi and Crutchfield 2001; Givan, Dean, and Greig 2003).

Self-closure is distinct from sufficiency for a specified prediction target. If the quantity of interest is

$$Z = g(X_{t+1}),$$

the required condition is that

$$P(X_t) \longrightarrow Z$$

be single-valued. It is not necessary for

$$P(X_t) \longrightarrow P(X_{t+1})$$

to be single-valued.

This distinction appears directly in the finite ABCW dataset. For the field-only observation

$$P_W(X) = \Delta W,$$

the next field is unique for 1,390 of the 2,562 observed current fields:

$$\frac{1390}{2562} \simeq 0.54254.$$

Thus, the current field alone determines the complete next field for approximately 54.3% of the observed current fields, but not in general.

When the current action is retained through

$$P_{aW}(X) = (a, \Delta W),$$

the relation

$$(a_t, \Delta W_t) \longrightarrow \Delta W_{t+1}$$

is single-valued for all 11,202 observed $(a, \Delta W)$ states. By contrast, the next state of this observation itself,

$$(a_t, \Delta W_t) \longrightarrow (a_{t+1}, \Delta W_{t+1}),$$

is single-valued for only 8,390 of the 11,202 observed states:

$$\frac{8390}{11202} \simeq 0.748973.$$

This is approximately 74.9%. An observation that is not self-closed can therefore still predict a specified component of the future perfectly. We keep these two requirements separate and fix the prediction target explicitly below.

One-Step Exact-Field Prediction

The prediction target in this paper is

$$Z = \Delta W_{t+1}.$$

That is, the observation must predict the **complete one-step-ahead field deviation** without error.

Retaining $(a_t, \Delta W_t)$ makes this target single-valued throughout the finite dataset, but it also retains all 2,562 distinct field deviations. We therefore keep the current action a_t unchanged while further coarsening only the field component.

Let

$$B : \Delta W \longrightarrow \mathcal{C}$$

be a field-only map, and define the corresponding observation as

$$P_B(X_t) = (a_t, B(\Delta W_t)).$$

The map B satisfies the predictive requirement if, for any two current states X_t and X'_t in the analyzed transitions,

$$a_t = a'_t$$

and

$$B(\Delta W_t) = B(\Delta W'_t)$$

imply

$$\Delta W_{t+1} = \Delta W'_{t+1}.$$

Equivalently, we require

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

to be single-valued on the stored finite transition set.

The action a is not itself compressed, and the prediction target is not coarse-grained: the required output remains the complete next field deviation. The problem considered here is therefore an **action-conditioned, one-step, exact-output, field-only predictive-partition problem**.

We call the required single-valuedness **predictive sufficiency** for the specified target. In the present formulation, B is admissible when $(a_t, B(\Delta W_t))$ is predictively sufficient for ΔW_{t+1} on the analyzed dataset.

Minimum Field Partition

Field deviations mapped to the same value by B are treated as members of the same class, and the set of classes is $\text{Im } B$. The complete field observation has

$$|\text{Im } B| = 2562.$$

If exact predictive accuracy can be retained with

$$|\text{Im } B| < 2562,$$

then distinct field deviations can be identified for the specified prediction task, at least on the finite reachable set considered here.

We define the optimization problem as

$$\min_B |\text{Im } B|$$

subject to

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1} \quad \text{being single-valued on all observed transitions.}$$

This formulation is related in motivation to the Information Bottleneck, which compresses a representation while retaining relevant information. It is not the standard Information Bottleneck objective. Rather than optimizing a trade-off between compression and relevant information through mutual information, we impose zero prediction error as a hard constraint and minimize the finite-set class count $|\text{Im } B|$ (Tishby, Pereira, and Bialek 2000).

The problem also differs from finding the coarsest self-closed partition. Here, the retained action serves as a conditioning variable, and the quantity preserved is the external target ΔW_{t+1} . The minimization is therefore defined by target prediction rather than self-closure of the observation.

When useful, we report the field-class compression rate

$$C_{\Delta W} = 1 - \frac{|\text{Im } B|}{2562}.$$

This is a class-count-based compression rate. It is not a reduction rate in Shannon information or in the number of bits required for representation.

This formulation separates the minimization problem from the prior selection of features that appear intuitively natural. Section 5 first evaluates baselines constructed from natural network features. Section 6 then derives the required field distinctions directly from the predictive constraint and determines the minimum partition.

Natural-Feature Baselines

Before deriving the minimum predictive partition directly from the dynamics, we examine how coarsely the current field ΔW can be represented by human-selected network features, including norms, local structure, hub structure, and out-strength. The purpose is to assess whether interpretability, genuine field compression, and predictive sufficiency for the complete one-step-ahead field can be achieved simultaneously.

Natural-feature compression and its limit

The out-strength vector, for example, achieves very high predictive performance. Its state-level determinism score is evaluated over the observed states defined by $(a, \text{out-strength})$. The next field is unique for 11,084 of the 11,142 distinct observed states:

$$\frac{11084}{11142} = 0.994794\dots$$

The resulting state-level determinism score is 99.4794%. The remaining 58 observed states are conflicts: more than one next field occurs from the same observed state.

The denominator here is not the total number of 56,536 transitions but the 11,142 distinct observed states. This state-level score should therefore be distinguished from evaluations weighted over the complete transition set.

We then searched for simple feature combinations that would resolve these 58 conflicts while genuinely compressing ΔW . Within the candidate family examined, no feature representation simultaneously achieved nontrivial field compression and exact predictive determinism.

This result contrasts with the response-based minimum predictive partition constructed exactly in Section 6. That partition compresses 2,562 fields into 692 classes while preserving exact one-step field prediction across all 56,536 observed transitions.

The contrast reflects two different approaches to compression. The first summarizes the field using features selected in advance:

$$\Delta W \longrightarrow B(\Delta W).$$

The second derives the necessary distinctions from equivalence of future responses:

$$\Delta W \longrightarrow (a \longmapsto \Delta W').$$

The first approach favors interpretability but may discard information required for prediction. The second preserves distinctions required by the predictive task, but the resulting classes need not have a simple interpretation as network statistics. The 692-class partition derived in Section 6 belongs to the second category.

Exhaustive combinations of the candidate feature families

The exhaustive search uses the following nine feature maps. For a field matrix $D = \Delta W$ with singular values $\sigma_1(D) \geq \dots \geq \sigma_5(D) \geq 0$, define its ordered out- and in-strength vectors by

$$r(D) = D\mathbf{1}, \quad c(D) = D^\top \mathbf{1}.$$

The nine maps are:

1. **Squared Frobenius norm:** $\|D\|_F^2 = \sum_{i,j} D_{ij}^2$.
2. **Matrix rank:** $\text{rank}(D)$, evaluated numerically on the integer-valued matrix.
3. **Stable rank:** $\text{srnk}(D) = \|D\|_F^2 / \sigma_1(D)^2$, with value zero when $\sigma_1(D) = 0$.
4. **Singular-value vector:** $(\sigma_1(D), \dots, \sigma_5(D))$.
5. **Out-strength distribution:** the entries of $r(D)$ sorted in nondecreasing order.
6. **In-strength distribution:** the entries of $c(D)$ sorted in nondecreasing order.
7. **Joint in/out-strength distribution:** the five pairs $(c_i(D), r_i(D))$ sorted lexicographically.
8. **Out-strength vector:** the ordered vector $r(D)$, retaining agent identities.
9. **In-strength vector:** the ordered vector $c(D)$, retaining agent identities.

Thus, “distribution” denotes a permutation-invariant sorted representation, whereas “vector” preserves the ordering of the five agents. Numerically computed singular values and stable rank are canonicalized to ten significant digits in the exhaustive comparison; the remaining feature values are integer-valued. A feature combination is the tuple of its selected map values, with no learned weighting or post-processing.

These nine feature maps have

$$2^9 - 1 = 511$$

nonempty subsets. We exhaustively evaluated all 511 feature combinations. None simultaneously merged genuinely distinct values of ΔW and determined the complete one-step-ahead field without ambiguity.

The out-strength vector yields high determinism but leaves 58 conflicting observed states. A feature representation denoted by B^* achieves exact prediction, but it uniquely distinguishes all 2,562 fields and therefore provides no genuine field compression.

Table 3 compares representative natural-feature baselines with the exact minimum partition. Here D_{state} is the state-level determinism score, whereas A_{freq} is the corresponding transition-frequency-weighted score.

Table 3. Natural-feature baselines and exact minimum partition

Observation	Field classes	Observed (a, B) states	D_{state}	A_{freq}	True field compression
Full field ΔW	2562	11202	1	1	No
Out-strength vector	2516	11142	0.994794	0.994658	Yes

Observation	Field classes	Observed (a, B) states	D_{state}	A_{freq}	True field compression
Best true-compression candidate by D_{state} (in-strength distribution + out-strength vector)	2561	11200	0.999821	0.999717	Yes
B^* (Frobenius norm + out-strength vector)	2562	11202	1	1	No
Exact minimum partition B_{692}	692	8084	1	1	Yes

Figure 3 places all 511 candidates in the same performance–compression plane and highlights the exact minimum partition.

Figure 3. Prediction performance versus actual field compression. All 511 nonempty combinations from the nine natural feature families are plotted using the number of distinct field values and the state-level determinism score. The out-strength vector, the 100%-predictive but noncompressive B^ representation, the best genuinely compressive candidate by D_{state} , and the exact 692-class partition are highlighted.*

The failure of the natural-feature search to achieve both exact prediction and genuine compression does not imply that nearly all 2,562 current fields must remain distinct. It shows only that the tested feature family does not express the required partition. In the next section, we remove the restriction of selecting features in advance and derive the necessary distinctions directly from conflicts in future responses.

Exact Minimum Predictive Partition

Incompatibility graph

We now remove the restriction of preselected features and derive directly from the dynamics which distinctions must be retained to avoid confusing future fields. The analyzed finite reachable set is the same as in Sections 3–5 and contains:

- 56,536 transitions;
- 2,562 distinct current fields; and
- 11,202 distinct $(a, \Delta W)$ states.

Consider a field-only map

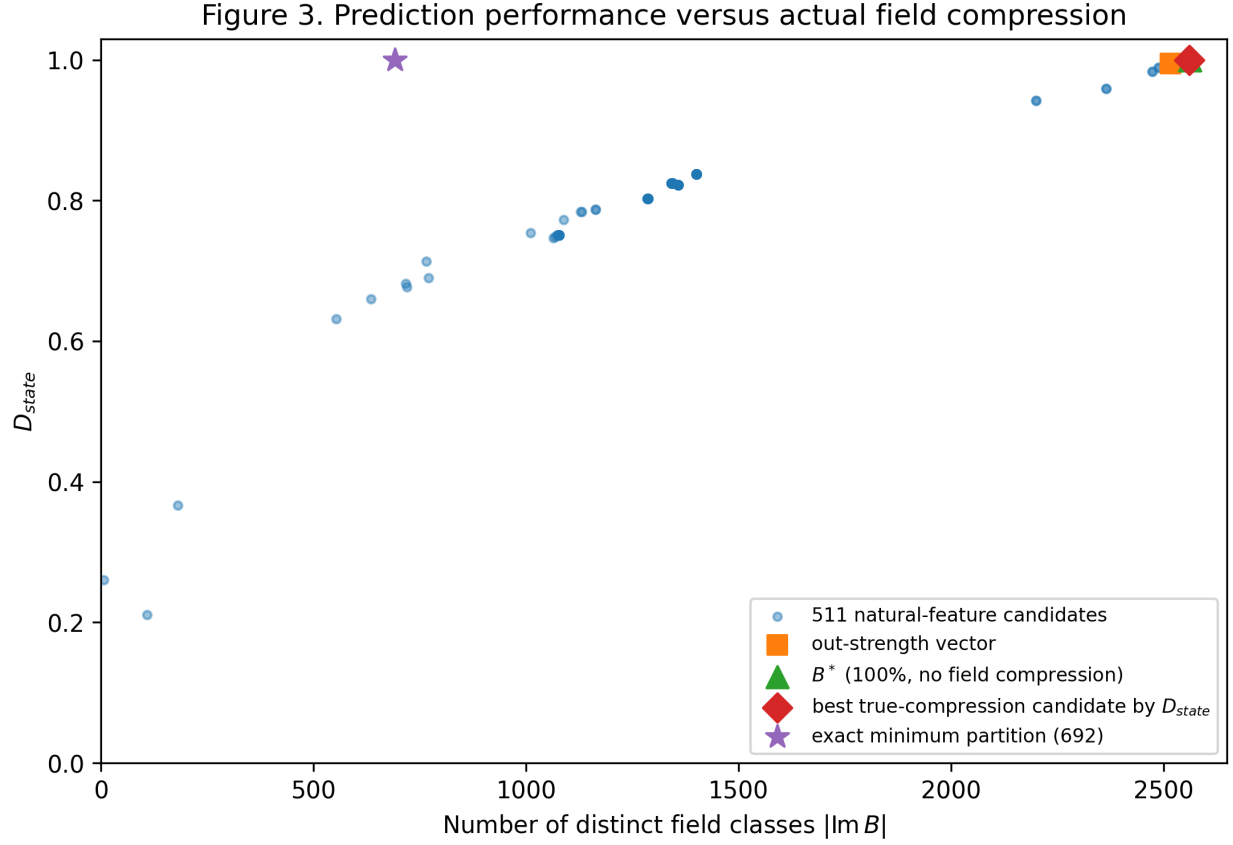


Figure 3: Figure 3. Prediction performance versus actual field compression. All 511 nonempty combinations from the nine natural feature families are plotted using the number of distinct field values and the state-level determinism score. The out-strength vector, the 100%-predictive but noncompressive B^* representation, the best genuinely compressive candidate by D_{state} , and the exact 692-class partition are highlighted.

$$B : \Delta W \longrightarrow \mathcal{C}$$

and the observation

$$P_B(X_t) = (a_t, B(\Delta W_t)).$$

The requirement is that

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

be single-valued over all 56,536 observed transitions. In other words, the same action and the same field class must never produce different next fields.

Consider two current fields ΔW_i and ΔW_j . If both are observed under a common action a and produce different next fields $\Delta W'_i \neq \Delta W'_j$, they cannot be assigned to the same field class. If

$$B(\Delta W_i) = B(\Delta W_j),$$

then the identical observation

$$(a, B(\Delta W_i)) = (a, B(\Delta W_j))$$

would lead to two different next fields, violating single-valued exact prediction.

We therefore construct an incompatibility graph G with the 2,562 distinct current fields as vertices. An edge joins two fields if they produce different next fields under at least one common observed action. A field-only partition preserving exact prediction is then equivalent to a proper coloring of G , because adjacent vertices must receive different class labels.

It follows that the minimum number of field classes is

$$|\text{Im } B_{\min}| = \chi(G),$$

where $\chi(G)$ is the chromatic number of the incompatibility graph.

This formulation does not introduce a new general state-minimization problem. It is connected to compatibility-based minimization of partially specified input-output behavior, particularly a depth-1 partial Mealy-machine formulation of incompletely specified finite-state machine minimization. The objective here is to obtain the concrete exact solution produced when this established problem form is applied to the finite reachable dynamics of ABCW.

Constructive upper bound

We first color the incompatibility graph using DSATUR (Br  laz 1979). This produces a valid proper coloring with 692 colors and hence establishes

$$\chi(G) \leq 692.$$

The number returned by a coloring algorithm is not, by itself, sufficient to verify the original prediction requirement. We therefore interpret the resulting coloring as the field-only map

$$B_{692}(\Delta W)$$

and map it back to all 56,536 original transitions. We then directly test whether

$$(a, B_{692}(\Delta W)) \longrightarrow \Delta W_{t+1}$$

is single-valued. The result is:

- field classes: 692;
- distinct (a, B_{692}) states: 8,084;
- nondeterministic observed states: 0; and
- one-step complete-field prediction: 100%.

Thus, a field-only partition exists that compresses

$$2562 \longrightarrow 692$$

while preserving exact one-step prediction of the complete field. The class-count-based compression rate is

$$C_{\Delta W} = 1 - \frac{692}{2562} \simeq 0.7299,$$

or approximately 72.99%. The 11,202 complete $(a, \Delta W)$ observations are reduced to 8,084 distinct $(a, B_{692}(\Delta W))$ observations.

Therefore, 692 is an upper bound:

$$\boxed{\chi(G) \leq 692}.$$

At this stage, however, a proper coloring with 691 or fewer colors has not been ruled out. Establishing exact minimality requires an independent lower bound of at least 692.

Anchor lower bound

To construct a lower bound, we fix one action condition. We choose the anchor action

$$a = (-, -, -, -, -).$$

Under this action, 1,239 current fields are observed, and they produce 625 distinct next fields. The lower-bound argument does not assume that these anchor groups cover all 2,562 fields. It uses only the induced subgraph on the 1,239 vertices actually observed under the anchor action.

Two current fields producing different next fields under the same anchor action cannot belong to the same field class. Partition the 1,239 fields into 625 anchor future groups

$$V_1, \dots, V_{625}$$

according to their next field under the anchor action. Every pair of vertices belonging to different groups is adjacent in G .

Lemma 1 — Additivity across anchor groups

For two distinct anchor future groups V_i and V_j , with $i \neq j$, let

$$u \in V_i, \quad v \in V_j.$$

By construction, u and v produce different next fields under the anchor action. Therefore,

$$\{u, v\} \in E(G),$$

and every two distinct anchor future groups form a complete join. Consequently, different anchor groups cannot share a color in any proper coloring.

Let

$$G_i = G[V_i]$$

be the subgraph induced by group V_i . The chromatic number of the anchor-observed induced subgraph is then additive:

$$\chi\left(G\left[\bigcup_{i=1}^{625} V_i\right]\right) = \sum_{i=1}^{625} \chi(G_i).$$

Because every group requires at least one color,

$$\chi(G) \geq 625.$$

Combining this with the constructive upper bound gives

$$\boxed{625 \leq \chi(G) \leq 692}.$$

The value 625 captures only the distinctions exposed by the anchor action. Two fields that lead to the same anchor future may still lead to different futures under another action condition. Such fields must be separated when all observed actions are considered simultaneously.

We therefore inspect the incompatibilities remaining within each of the 625 anchor future groups. The result is:

- 575 groups with no internal conflicts; and
- 50 groups with internal conflicts.

For the 575 conflict-free groups, $\chi(G_i) = 1$. The chromatic numbers of the remaining 50 induced subgraphs must be determined exactly.

Figure 6. Global sign-flip symmetry

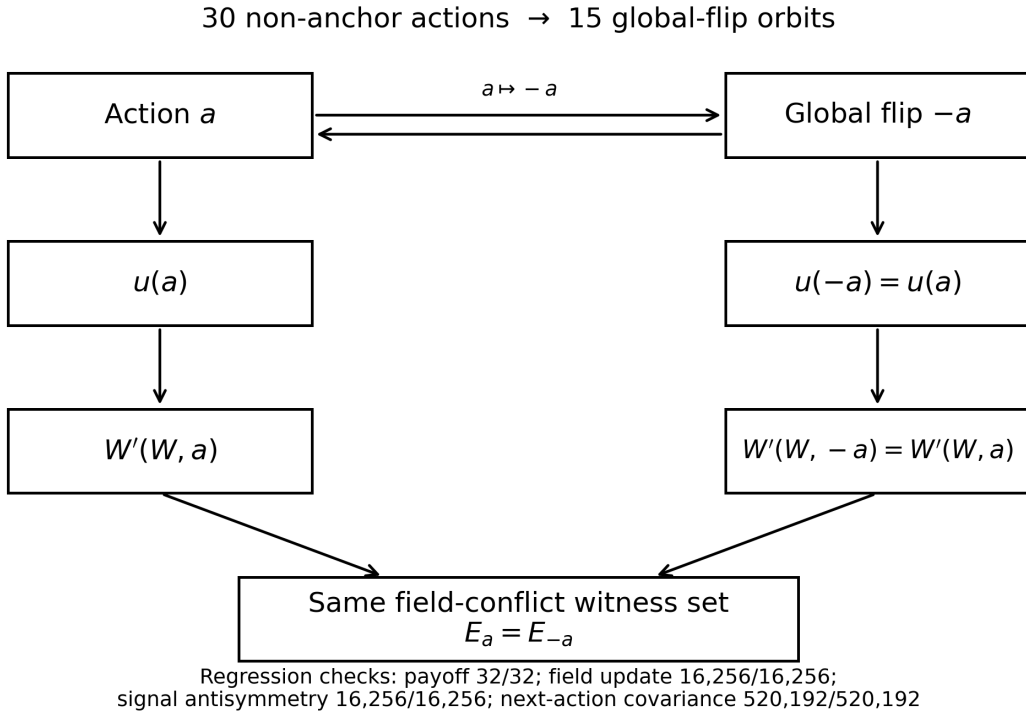


Figure 4: Figure 4. Global sign-flip symmetry. Under the global action flip $a \rightarrow -a$, minority-game payoffs are unchanged and the one-step field update is invariant, so a and $-a$ witness exactly the same field-conflict edges. Hence 30 non-anchor actions collapse to 15 global-flip orbits.

Figure 4. Global sign-flip symmetry. Under the global action flip $a \rightarrow -a$, minority-game payoffs are unchanged and the one-step field update is invariant, so a and $-a$ witness exactly the same field-conflict edges. Hence 30 non-anchor actions collapse to 15 global-flip orbits.

Under this symmetry, actions a and $-a$ witness the same set of field-conflict edges. The 30 non-anchor actions can therefore be grouped into 15 global-flip orbits for the purpose of describing internal conflicts. This symmetry does not itself provide the lower bound of 692. It identifies duplicated conflict information and supplies both a redundancy reduction and a consistency check for interpreting the internal analysis of the 50 difficult groups.

Exact refinement of the 50 difficult anchor groups

For each of the 50 anchor groups containing internal conflicts, we compute the chromatic number of the induced incompatibility graph G_i exactly.

For each group, a greedy DSATUR coloring provides an upper bound (Brélaz 1979), and a Bron–Kerbosch-type maximum-clique search provides a lower bound (Bron and Kerbosch 1973). If the bounds coincide, their common value determines $\chi(G_i)$ immediately. If they do not, we test k -colorability by backtracking in DSATUR order for successive values of k from the lower bound to the upper bound. The smallest feasible value is the exact chromatic number.

The values reported below are therefore not merely the color counts returned by a greedy heuristic. For every induced subgraph, they are certified by matching lower bounds and colorability results.

Across all 625 anchor groups, the exact chromatic-number distribution is:

- $\chi = 1$: 575 groups;
- $\chi = 2$: 39 groups;
- $\chi = 3$: 8 groups;
- $\chi = 4$: 2 groups; and
- $\chi = 7$: 1 group.

The group requiring the largest number of colors, $\chi = 7$, has 44 vertices and 224 internal conflict edges. Its maximum-clique lower bound and DSATUR upper bound are both 7, directly establishing

$$\chi(G_i) = 7.$$

The vertex count, internal edge count, exact chromatic number, lower bound, and upper bound for every group are stored in the reproducibility data.

By Lemma 1, colors cannot be shared across distinct anchor future groups. The total number of colors required by the anchor-observed induced subgraph is therefore the sum of the group chromatic numbers. Relative to assigning one color to each of the 625 groups, the additional number of colors is

$$39(2 - 1) + 8(3 - 1) + 2(4 - 1) + 1(7 - 1)$$

$$= 39 + 16 + 6 + 6$$

$$= 67.$$

Hence,

$$625 + 67 = 692.$$

The same result is obtained by the direct cross-check

$$575 \cdot 1 + 39 \cdot 2 + 8 \cdot 3 + 2 \cdot 4 + 1 \cdot 7 = 692.$$

Thus, the induced subgraph on the 1,239 fields observed under the anchor action already requires 692 colors. The complete incompatibility graph must therefore satisfy

$$\chi(G) \geq 692.$$

The value 692 is consequently not an accidental output of DSATUR. The anchor action alone forces 625 distinctions. The remaining 67 arise because some fields that are indistinguishable under the anchor condition must be separated under other action conditions. Moreover, these additional distinctions are localized in only 50 of the 625 groups.

The decomposition

$$692 = 625 + 67$$

therefore reflects the complete-join structure across anchor groups and the action-conditioned conflicts remaining within them. Figure 5 summarizes the matching upper and lower bounds and this decomposition.

Table 4 records the exact contribution of each anchor-group chromatic number to the lower bound.

Table 4. Exact lower-bound decomposition

Required colors per anchor group	Number of anchor groups	Contribution to total classes	Additional colors beyond one per group
1	575	575	0
2	39	78	39
3	8	24	16
4	2	8	6
7	1	7	6
Total	625	692	67

Figure 5. Exact minimum predictive partition. The constructive 692-coloring provides the upper bound. The anchor action yields 625 groups; exact coloring of the 50 internally conflicting groups requires 67 additional colors, producing a matching lower bound of 692.

Main result

Section 6.2 constructs a valid 692-coloring of the incompatibility graph on all 2,562 fields, establishing

$$\chi(G) \leq 692.$$

Direct evaluation of this coloring on all 56,536 original transitions verifies that

$$(a, B_{692}(\Delta W)) \longrightarrow \Delta W_{t+1}$$

is single-valued throughout the observed dataset.

Figure 4. Exact minimum predictive partition: matching upper and lower bounds

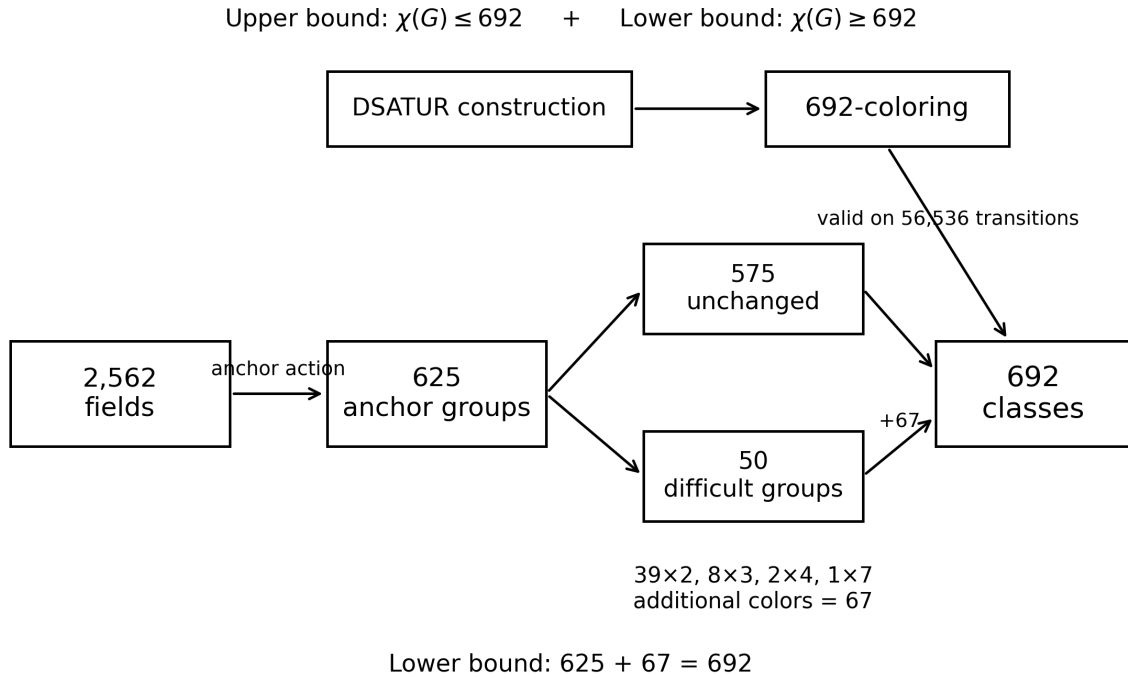


Figure 5: Figure 5. Exact minimum predictive partition. The constructive 692-coloring provides the upper bound. The anchor action yields 625 groups; exact coloring of the 50 internally conflicting groups requires 67 additional colors, producing a matching lower bound of 692.

Sections 6.3 and 6.4 decompose the 1,239 fields observed under the anchor action into 625 future groups. The complete joins across different groups and the exact chromatic numbers within the groups establish

$$\chi(G) \geq 692.$$

The bounds coincide:

$$\boxed{\chi(G) = 692}.$$

Main Result.

For the 56,536 observed transitions and 2,562 distinct current fields of the five-agent ABCW system, consider a field-only map

$$B : \Delta W \longrightarrow \mathcal{C}$$

such that

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

is single-valued over all observed transitions. The exact minimum number of field classes is

$$\boxed{|\text{Im } B_{\min}| = 692}.$$

It is therefore unnecessary to retain all 2,562 distinct current fields for the specified task. Exact prediction of the complete one-step-ahead field permits the reduction

$$2562 \longrightarrow 692,$$

with the class-count-based compression rate

$$1 - \frac{692}{2562} \simeq 72.99\%.$$

Any partition into 691 or fewer classes fails even on the anchor-induced subgraph used in the lower-bound argument. It must assign the same class to at least one pair of fields that produce different next fields under a common action condition, thereby violating exact prediction.

Thus, 692 is not merely the size of a high-performing candidate partition. It is the minimum number of classes in a field-only partition that preserves exact one-step, action-conditioned complete-field prediction on the observed finite reachable set.

The result also clarifies the contrast with the natural-feature baselines. Among the 511 tested feature combinations, none achieved both genuine field compression and exact prediction. This was not because exact prediction intrinsically required nearly complete microscopic field identification. Once the restriction to preselected features is removed and the partition is derived from future incompatibilities, the 2,562 fields can be reduced to 692 classes. Hence,

$$\boxed{\text{human-designed features} \neq \text{minimum distinctions required for prediction}}.$$

The number 692 is not a universal state count for ABCW. It is the exact solution under the following conditions:

- five agents;
- the finite reachable set generated in this study;
- the 56,536 observed transitions;
- field-only compression;
- retention of the action a ;
- one-step prediction; and
- the complete ΔW_{t+1} as the prediction target.

Different initial conditions, reachable sets, numbers of agents, prediction horizons, or coarser targets Q may produce different minimum partitions.

The theorem determines the chromatic number, and hence the minimum class count of 692. The concrete map B_{692} verified above is one representative minimum partition obtained from the proper coloring constructed by DSATUR. A different 692-coloring—including one constructed from the anchor decomposition—may assign fields to classes differently. We therefore claim uniqueness only of the minimum class count for this incompatibility graph, not uniqueness of the minimizing map or partition.

The search for the minimum number of classes is now complete. Determining that the reduction is $2562 \rightarrow 692$, however, is not the same as understanding what the 692 classes represent. Section 7 therefore examines which fields are merged within the classes, which distinctions are exposed by action conditions, how the additional 67 distinctions are localized, and what symmetries are present in the partition structure.

Structure and Interpretation of the Minimum Partition

The preceding analysis established that 692 is the minimum number of classes required to predict the next field $\Delta W'$ uniquely from the current field ΔW conditioned on the action a , over the 2,562 observed fields.

This result is not merely a statement that 2,562 states can be compressed into 692 states. We now examine the structure of the 692 classes, why 625 classes are insufficient, and why 67 additional distinctions are required.

The objective is not to assign an immediately interpretable meaning to every class. Rather, it is to clarify how the distinctions required to preserve the future relate to distinctions among the original microscopic fields.

The minimum as a predictive distinction count

The partition obtained here is not an arbitrary clustering of the observed fields ΔW . Two fields ΔW_i and ΔW_j can be assigned to the same class only if identifying them preserves unique prediction of the next field under all observed action conditions shared by the two fields.

Among maps of the form

$$B : \Delta W \longrightarrow \{1, \dots, K\},$$

we require

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

to be single-valued on the observed dataset. The exact minimum is

$$K_{\min} = 692.$$

No universal meaning should be assigned to the number 692 itself. It is determined by the five-agent ABCW model, the four initial fields, the finite reachable set generated from them, and the target of predicting the complete one-step-ahead field ΔW_{t+1} . We therefore refer to the result as the minimum predictive partition preserving one-step, action-conditioned field responses on this dataset, rather than as a universal number of ABCW macrostates.

Class-size structure and genuine merging

The original observation set contains 2,562 distinct values of ΔW . The identity observation

$$P_{\text{id}}(\Delta W) = \Delta W$$

distinguishes all of them and therefore has 2,562 classes. By contrast, the minimum predictive partition reduces the distinction count to

$$2562 \longrightarrow 692.$$

In terms of field-class count, approximately 73% of the original distinctions can be discarded without losing one-step field prediction on the observed dataset.

This compression is not produced solely by a small number of exceptionally large classes. Of the 692 classes:

- 352 contain a single ΔW ;
- 340 contain multiple distinct values of ΔW ; and
- the largest class contains 65 distinct values of ΔW .

Many fields must therefore remain individually distinguished, but genuine merging is also widespread. Microscopically different fields can often be treated as identical for the specified prediction task. The result provides a concrete finite example in which

microscopically distinct

does not imply

must remain distinct for prediction.

The 625-class anchor decomposition

The intermediate lower bound of 625 is central to the internal structure of the exact result. It is obtained by fixing the anchor action

$$a = (-, -, -, -, -).$$

Under this action, 1,239 current fields are observed and produce 625 distinct next fields. Grouping current fields by their next field under the anchor action gives an initial decomposition into 625 anchor future groups.

This 625-group decomposition is not yet a valid complete predictive partition when the other observed action conditions are included. Of the 625 groups, 575 require no further subdivision. All remaining conflicts are localized in the other 50 groups:

$$625 = 575 + 50.$$

The increase from 625 to 692 is therefore not distributed uniformly across the partition.

The additional 67 distinctions

Refining the 50 difficult groups so that future fields remain unique under every observed action condition requires 67 additional distinctions:

$$625 + 67 = 692.$$

After refinement, the 50 original groups split into

$$39 \times 2 + 8 \times 3 + 2 \times 4 + 1 \times 7 = 117$$

subclasses. The increase is therefore

$$117 - 50 = 67.$$

Most predictive distinctions are already forced at the 625-group stage, whereas the remaining ambiguity is concentrated in a comparatively small subset. The minimum partition thus has the form

large base partition + localized additional refinement.

This structure is not apparent if the partition is viewed only as a lookup table containing 692 labels. Figure 6 summarizes both the final composition of 352 singleton and 340 multi-field classes and the localization of all 67 additional distinctions within 50 of the 625 anchor groups.

Figure 6. Internal structure of the 692-class partition. Top: composition of the 692 predictive classes (352 singleton and 340 multi-field classes), with verified mean, median, and maximum class size. Bottom: localization of all additional refinements to 50 of the 625 anchor groups.

Figure 5. Internal structure of the 692-class partition

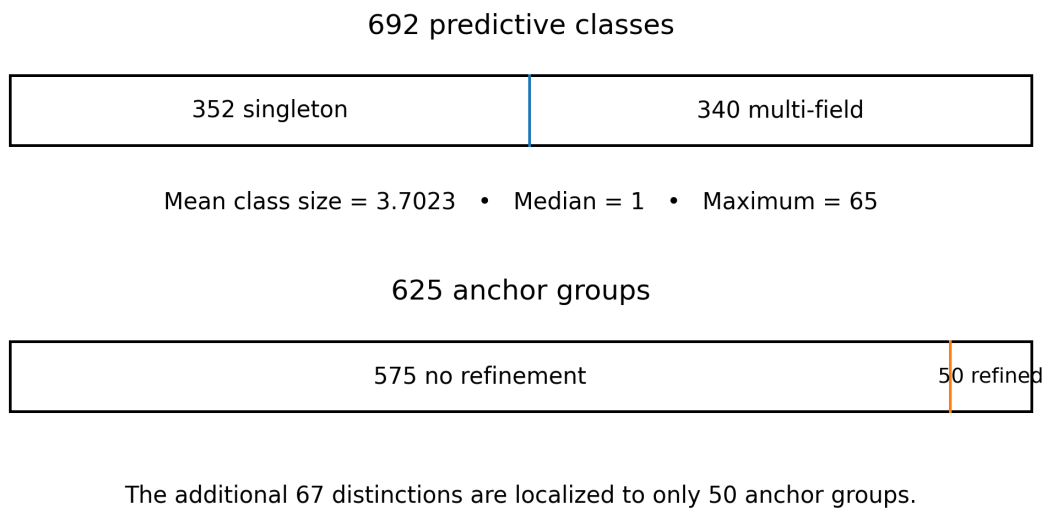


Figure 6: Figure 6. Internal structure of the 692-class partition. Top: composition of the 692 predictive classes (352 singleton and 340 multi-field classes), with verified mean, median, and maximum class size. Bottom: localization of all additional refinements to 50 of the 625 anchor groups.

Geometric proximity versus predictive equivalence

We next ask whether fields assigned to the same class are also similar in an ordinary geometric sense. If the minimum predictive partition resembled a simple geometric clustering, nearby values of ΔW would be expected to share classes, while distant fields would tend to be separated.

Inspection of the partition does not support this expectation in general. Fields that differ substantially in their microscopic representations can belong to the same predictive class, while nearby fields can be assigned to different classes.

The partition therefore does not preserve the geometry of the ΔW space itself. It preserves the action-conditioned future-response structure

$$\Delta W_t \mapsto (a \mapsto \Delta W_{t+1}).$$

In this predictive sense, two fields are “close” not because their matrix entries differ only slightly, but because they produce indistinguishable future responses under the relevant observed action conditions.

Predictive sufficiency and representation cost

These results make concrete the relationship between predictive sufficiency and compression. The identity observation P_{id} necessarily retains the information required for prediction, but it distinguishes all 2,562 fields and performs no compression. At the opposite extreme, a very coarse observation may compress the state description substantially while failing to distinguish the relevant futures.

The question is therefore:

How many distinctions can be discarded without losing the specified future?

For the finite dataset and one-step field-prediction target considered here, the boundary occurs at 692 classes. Compression and predictive sufficiency must be treated as separate axes: coarseness alone is insufficient, while predictability alone is trivially achieved by the identity observation. The objective is to retain only the distinctions required for prediction and discard the rest.

Why the partition is not yet an interpretable macrovariable

The existence of a 692-class minimum partition does not establish that a natural macrovariable of ABCW has been discovered.

First, the partition is constructed from an observed finite reachable set. Second, its target is restricted to the one-step-ahead field ΔW_{t+1} . Third, the classes are defined by future responses, and it has not been shown that they can be expressed through simple physical or network quantities. Fourth, exact determination of the minimum class count does not imply that the class labels or the partition realizing that count are unique: the DSATUR-derived B_{692} is a verified representative optimum, not a canonical quotient.

The result directly supports only the following claim:

In the observed finite ABCW system, not all 2,562 microscopic field distinctions are required to preserve exact one-step, action-conditioned field responses; 692 predictive distinctions are necessary and sufficient.

An intermediate descriptive level

The 692 classes occupy an intermediate descriptive level between a complete microscopic representation and a small set of simple macroscopic statistics. They retain the distinctions required for a specified future while discarding distinctions that are irrelevant to that target on the observed dataset.

They should not, however, be treated as natural macroscopic variables of ABCW in general. Their stability under larger systems, different initial conditions, and longer prediction horizons remains unresolved, as does the possibility of representing the 692 classes through a concise set of network observables. Section 8 discusses these implications and limitations.

Discussion

What the exact minimum means

The central question of this study was:

Which distinctions in the present must be retained in order to predict the future?

The finite reachable set observed for the five-agent ABCW model contains 2,562 distinct current fields ΔW . An identity field representation that distinguishes all of them necessarily retains the information required to predict the complete one-step-ahead field ΔW_{t+1} . The relevant question, however, is not whether the complete current field is sufficient, but how many of its distinctions can be discarded without losing the specified future.

The exact result obtained in Section 6 is

$$|\text{Im } B_{\min}| = \chi(G) = 692.$$

Under the conditions of the observed finite reachable set, field-only compression, retention of the current action a_t , and exact prediction of the complete next field ΔW_{t+1} , the current-field distinction count can be reduced as

$$2562 \longrightarrow 692.$$

With 691 or fewer classes, at least one pair of current fields producing different next fields under a common action condition must be identified, so exact prediction cannot be preserved.

The value 692 is therefore neither an arbitrary output of a compression algorithm nor a performance score for a chosen feature set. It is the minimum number of field distinctions that must be retained for the specified prediction task.

This interpretation has explicit limits. The value is not the state count of ABCW in general, nor is it a universal number of naturally occurring macrostates. It depends on the number of agents, initial fields, reachable set, observation form, prediction target, time horizon, and requirement of exact prediction. The main result is therefore most precisely stated as follows:

In the finite ABCW system examined here, all 2,562 microscopic field distinctions are unnecessary for predicting the complete one-step-ahead field, but at least 692 predictive field distinctions must be retained.

Microscopic difference and predictive relevance

The central interpretation is that

microscopically different

and

must remain distinct for the specified prediction

are not equivalent.

The minimum partition contains 352 singleton classes and 340 classes that merge multiple distinct fields, with the largest class containing 65 current fields. Many fields must remain individually identifiable, but many others can be merged despite being microscopically different.

Class membership also does not coincide with simple geometric proximity between field matrices. Fields that differ substantially can share a predictive class, whereas nearby fields must be separated whenever they produce different next fields under a common action condition.

The minimum partition therefore preserves not the static geometry of the current fields themselves, but their observed action-conditioned one-step response structure:

$$\Delta W_t \mapsto (a \mapsto \Delta W_{t+1}).$$

This demonstrates a limitation of evaluating the fineness of a state description solely through the geometry or dimensionality of the present state. Once a predictive objective is fixed, what matters is whether a present distinction appears as a distinction in the target future.

Hand-designed observables and dynamics-derived partitions

Before deriving the minimum partition, we evaluated field compression based on human-selected network features, including norms, local structure, and out-strength. These features are not uninformative. In particular, out-strength gives a unique next field for 11,084 of 11,142 observed $(a, \text{out-strength})$ states, corresponding to a state-level determinism score of 99.4794%.

Nevertheless, 58 conflicting observed states remain. Across all 511 nonempty combinations of the nine candidate feature families, none both merged genuinely distinct values of ΔW and predicted the complete next field exactly.

Viewed in isolation, this failure could suggest that exact prediction requires an almost complete representation of the current field. Section 6 shows otherwise. Once the prior restriction to selected features is removed and incompatibilities in future responses are used directly, the 2,562 fields can be compressed into 692 classes.

For this dataset,

human-designed features $\neq$ minimum distinctions required for prediction.

This does not support a general conclusion that natural features are unsuitable for prediction. The tested family is finite, and other features or constructions may express the 692-class partition, or another high-performing compression, concisely. The result instead demonstrates a complementary direction of analysis: first derive from the dynamics the distinctions required to preserve the future, and then seek an interpretable representation of those distinctions.

The 692-class partition is therefore better understood as a predictive benchmark for future searches for interpretable macroscopic descriptions than as a final set of interpretable macrovariables.

Relation to state aggregation and predictive-state ideas

As discussed in Section 2, the problem has connections to state aggregation, lumpability, bisimulation, computational mechanics, and minimization of incompletely specified finite-state machines.

The connection to lumpability lies in the shared objective of retaining a well-defined description of future behavior after microscopic states have been aggregated. In the present study, however, we distinguish self-closure,

$$P(X_t) \longrightarrow P(X_{t+1}),$$

from prediction of a specified target,

$$P(X_t) \longrightarrow Z.$$

The preserved target is the complete one-step-ahead field ΔW_{t+1} , not the next state of the reduced field class. The 692-class partition is therefore not shown to define closed reduced dynamics.

The study also shares with bisimulation and MDP minimization the principle of aggregating states according to future behavior. It does not, however, preserve an optimal policy, reward structure, or complete reduced model. The current action is retained, only the field variable is compressed, and the preserved output is the complete next field.

The conceptual connection to causal states in computational mechanics is especially close because both approaches retain distinctions according to their relevance for the future. Causal states classify histories by conditional distributions over future sequences, whereas the present construction partitions current fields in a finite reachable set and restricts the target to one time step. Moreover, the input-output data are partially specified: not every action condition is observed for every current field. We therefore do not identify the 692 classes with causal states or an ϵ -machine.

The most direct mathematical correspondence is with minimization of incompletely specified finite-state machines. The data can be viewed as the partially specified input-output relation

$$(\Delta W, a) \longmapsto \Delta W'.$$

Two current fields cannot share a class when they produce different outputs under the same observed input condition. This compatibility/incompatibility structure directly parallels the classical problem.

The present formulation does not require recursive consistency of successor classes, because the preserved target is the complete one-step-ahead field rather than a successor class label. Under this depth-1 restriction, the minimum field-class count is given directly by the chromatic number of the incompatibility graph.

The study therefore does not propose a new general theory of state minimization or predictive representation. Its specific contribution is to formulate this depth-1 predictive state-reduction problem for partially specified data generated by a finite competitive agent system and to solve the minimum exactly through a constructive upper bound and an independent matching lower bound.

The structure behind 692

The exact minimum is not produced by uniformly refining all 2,562 fields. Under the anchor action

$$a = (-, -, -, -, -),$$

the 1,239 observed current fields produce 625 distinct next fields, forcing an initial 625-group decomposition. When the other observed action conditions are considered, 575 groups contain no additional internal conflicts, while 50 require further refinement.

Exact refinement of these 50 groups produces 117 subclasses, increasing the total class count by

$$117 - 50 = 67.$$

Thus,

$$625 + 67 = 692.$$

The predictive distinctions are not distributed uniformly across current fields. For most anchor groups, the distinction exposed by a single action condition is sufficient. Additional action conditions require new distinctions only within 50 groups. The minimum partition can therefore be understood as

$$\text{large base partition} + \text{localized additional refinement}.$$

This localization does not establish a general law that predictive information is localized. It is an observed structural property of the present finite dataset and prediction task. Nevertheless, it shows that tracing which conditions expose new incompatibilities can reveal where predictively relevant differences occur, information that is hidden if the result is represented only as a lookup table of 692 labels.

Scope and limitations

The result has at least six important limitations.

First, the analysis concerns a five-agent ABCW model and the finite reachable set generated from four initial fields. The value 692 cannot be extrapolated directly to other numbers of agents, other initial conditions, or ABCW systems in general.

Second, the current action a_t is retained, and only the field variable ΔW_t is compressed. The study does not determine a general minimum state representation that compresses action and field jointly.

Third, the prediction target is the complete one-step-ahead field ΔW_{t+1} . We have not established that the 692 classes are sufficient for multistep prediction or that they support closed recursive dynamics.

Fourth, the analyzed input-output behavior is partially specified. Not every action condition is observed for every current field. The exact minimum therefore applies to the observed finite reachable set and does not guarantee predictions for unobserved combinations.

Fifth, exact one-step prediction is imposed as a hard constraint. This makes the minimum partition mathematically well defined, but it does not imply that exact prediction is the only scientifically meaningful objective. Simpler representations such as out-strength can provide high, though imperfect, predictive performance.

Sixth, establishing a minimum of 692 classes does not show that those classes can be expressed through a concise set of network features, an algebraic rule, or a small number of interpretable variables. Nor does the minimum class count imply uniqueness of the partition that attains it.

These limitations specify, rather than weaken, the scope of the main result. What has been established exactly is the minimum number of field classes necessary and sufficient under a specified finite dataset, observation form, prediction target, and time horizon.

Future directions

Several extensions follow naturally from these limitations.

The first is to extend the prediction horizon. Two fields that can be merged for one-step prediction may generate different futures over multiple time steps. Conversely, if the target is changed from the complete future field to a coarser quantity, a partition with fewer than 692 classes may suffice.

The observation class may also be generalized from field-only compression to a joint map

$$C : (a, \Delta W) \longrightarrow \mathcal{Z},$$

so that action distinctions and field distinctions are optimized simultaneously rather than retaining a as an uncompressed conditioning variable. For an h -step target, depth-1 compatibility is no longer sufficient: on the partially specified transition system, two records merged at depth h must agree on the required immediate output and, whenever both successors are specified, their successor records must remain compatible at depth $h - 1$. This gives the recursive compatibility condition familiar from incompletely specified FSM minimization and provides a direct route to finite-horizon extensions of the present graph construction.

The second is to move from exact to approximate prediction. This study imposes exact prediction and minimizes the resulting class count. In practical model analysis, the trade-offs among predictive performance, compression, and representation cost may be more informative. Many intermediate representations may exist between simple, highly predictive features and the exact 692-class partition. These three quantities should therefore be treated as separate axes:

predictive performance,

compression,

representation cost.

Studying their Pareto structure is a natural next step.

The third direction is to seek a readable representation of the 692-class partition. The present result shows that the distinctions required for prediction can be derived from the dynamics, but it remains unknown whether they can be re-expressed through a small set of interpretable network quantities.

The fourth is to vary the number of agents, the initial fields, and the reachable set. Such comparisons are required to determine which aspects of the observed structure reflect more general properties of ABCW and which are specific to the present finite dataset.

These extensions are outside the main result of this paper. What is solved here is the more limited depth-1 problem. That restriction makes it possible to turn the abstract question of which distinctions matter for prediction into a finite problem that can be inspected and solved exactly.

Conclusion

Main result

This study determined the minimum number of field classes required, on a finite reachable set of the five-agent ABCW model, to preserve exact prediction of the complete one-step-ahead field ΔW_{t+1} while retaining the current action a_t and compressing only the current field ΔW_t .

We constructed an incompatibility graph on the 2,562 observed current fields. A proper 692-coloring provides a constructive upper bound, while an independent lower bound derived from an anchor action requires the same number of colors. Hence,

$$|\text{Im } B_{\min}| = \chi(G) = 692.$$

Not all 2,562 microscopic field distinctions are therefore required for the specified prediction task. At the same time, no field-only partition with 691 or fewer classes can preserve exact prediction.

Interpretation and scope

The exact minimum is not produced by a uniform refinement of the field set. The anchor action forces 625 base groups, of which only 50 require further subdivision under other action conditions. These localized refinements add 67 classes,

$$625 + 67 = 692,$$

and the final partition contains 352 singleton classes and 340 multi-field classes.

The natural-feature baselines also clarify the significance of the result. Across 511 combinations of nine feature families, the out-strength vector achieved a state-level determinism score of 99.4794%,

but no tested candidate combined genuine field compression with exact prediction. In this dataset, the features that appeared natural in advance did not coincide with the minimum distinctions required to preserve the future.

The value 692 is not a universal number of macrostates for ABCW. It depends on the five-agent model, the finite reachable set examined here, retention of the action, field-only compression, and exact one-step field prediction. Determining the minimum class count also does not imply uniqueness of the minimum coloring or establish that the classes can be expressed through a concise analytic observation map.

Outlook

Natural extensions include multistep prediction, trade-offs between compression and approximate predictive performance, scaling with the number of agents and the choice of initial fields, and the search for a smaller set of interpretable structural quantities capable of representing the 692-class partition.

The most limited conclusion of this paper is that **being distinct as a state is not the same as needing to remain distinct for a specified prediction**. The 692-class minimum predictive partition of this finite ABCW system provides one exact answer to the question of how much of the present can be forgotten without losing the specified future.

Data and Code Availability

The complete transition dataset, source code, exact-coloring certificate, lower-bound reconstruction, natural-feature analysis, and automated verification scripts are available at <https://github.com/youque8/abcw-paper-reproducibility>. The repository release corresponding to this manuscript is version 1.1.0. The core numerical chain can be checked by running `python scripts/verify_all.py`; the full recomputation, including all 511 natural-feature combinations and exact re-solving of the anchor refinements, is available through `python scripts/verify_full.py`.

References

- Arthur, W. Brian. 1994. “Inductive Reasoning and Bounded Rationality.” *American Economic Review* 84 (2): 406–11.
- Br  laz, Daniel. 1979. “New Methods to Color the Vertices of a Graph.” *Communications of the ACM* 22 (4): 251–56. <https://doi.org/10.1145/359094.359101>.
- Bron, Coen, and Joep Kerbosch. 1973. “Algorithm 457: Finding All Cliques of an Undirected Graph.” *Communications of the ACM* 16 (9): 575–77. <https://doi.org/10.1145/362342.362367>.
- Challet, Damien, and Yi-Cheng Zhang. 1997. “Emergence of Cooperation and Organization in an Evolutionary Game.” *Physica A: Statistical Mechanics and Its Applications* 246 (3–4): 407–18. [https://doi.org/10.1016/S0378-4371\(97\)00419-6](https://doi.org/10.1016/S0378-4371(97)00419-6).
- Givan, Robert, Thomas Dean, and Matthew Greig. 2003. “Equivalence Notions and Model Minimization in Markov Decision Processes.” *Artificial Intelligence* 147 (1–2): 163–223. [https://doi.org/10.1016/S0004-3702\(02\)00376-4](https://doi.org/10.1016/S0004-3702(02)00376-4).
- Kemeny, John G., and J. Laurie Snell. 1960. *Finite Markov Chains*. Van Nostrand.

- Kohavi, Zvi, and Niraj K. Jha. 2009. *Switching and Finite Automata Theory*. 3rd ed. Cambridge University Press.
- Paull, Marvin C., and Stephen H. Unger. 1959. “Minimizing the Number of States in Incompletely Specified Sequential Switching Functions.” *IRE Transactions on Electronic Computers* EC-8 (3): 356–67. <https://doi.org/10.1109/TEC.1959.5222697>.
- Shalizi, Cosma Rohilla, and James P. Crutchfield. 2001. “Computational Mechanics: Pattern and Prediction, Structure and Simplicity.” *Journal of Statistical Physics* 104 (3–4): 817–79. <https://doi.org/10.1023/A:1010388907793>.
- Tishby, Naftali, Fernando C. Pereira, and William Bialek. 2000. “The Information Bottleneck Method.” <https://arxiv.org/abs/physics/0004057>.